

CSE 4128

Image Processing and Computer Vision Laboratory

Week 3: Histogram Equalization

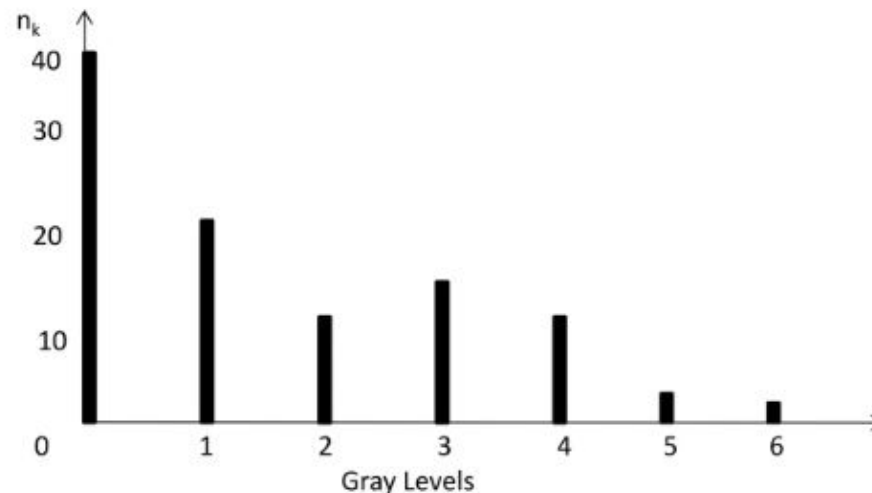
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Histogram

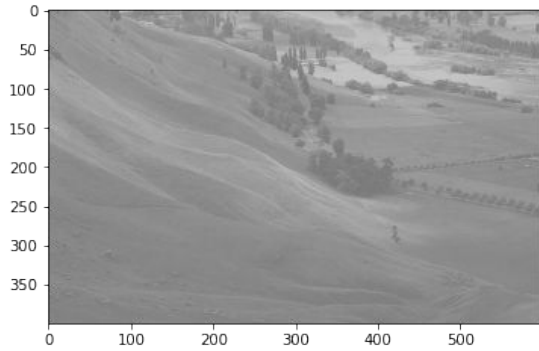
The histogram of an image consists of the x-axis representing the intensity levels r_k and the y-axis denoting the $h(r_k)$ or the $p(r_k)$ functions.

$$h(r_k) = n_k$$

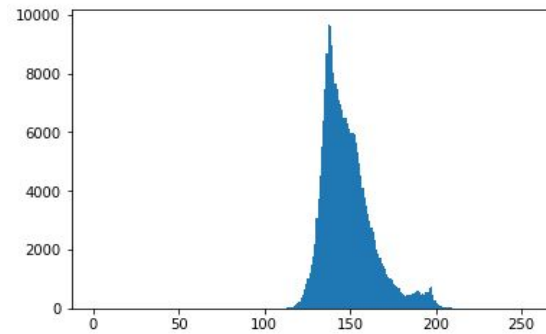


Histogram Equalization

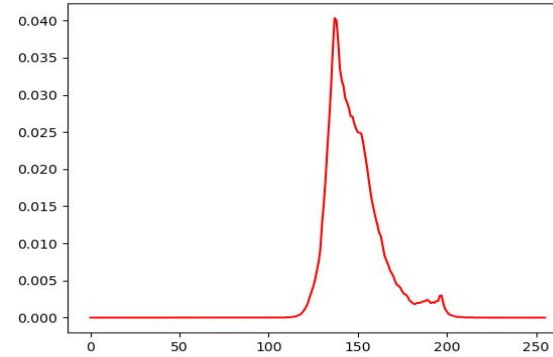
Histogram equalization is used to enhance contrast.



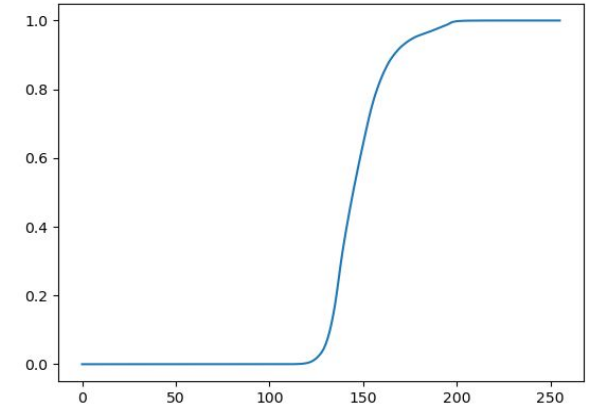
Input Image



Histogram of the Input Image



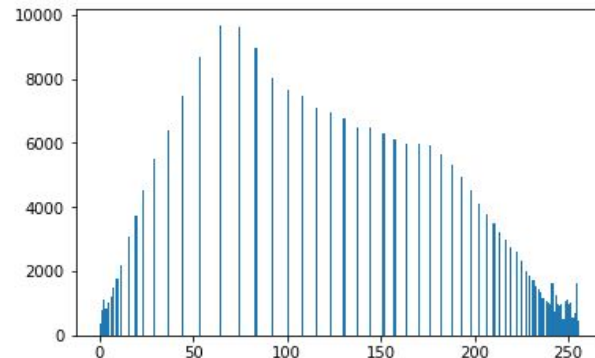
PDF of the Input Image



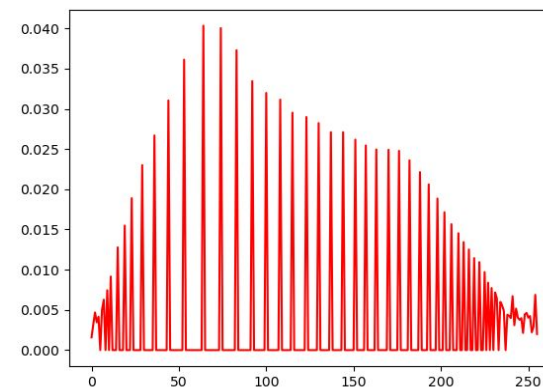
CDF of the Input Image



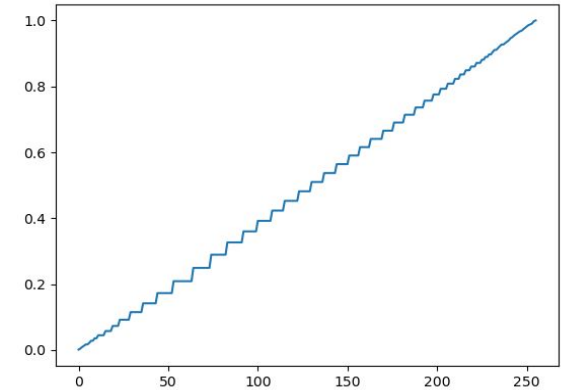
Equalized Image



Histogram of the Equalized Image



PDF of the Equalized Image



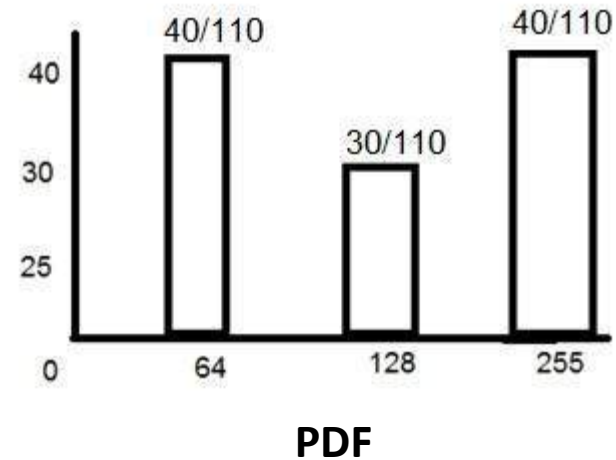
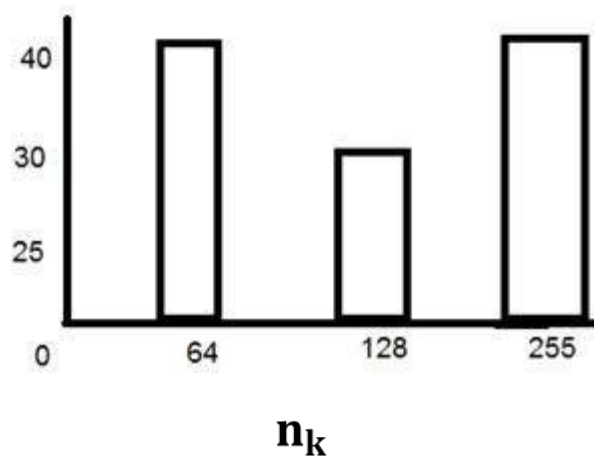
CDF of the Equalized Image

PDF

PDF stands for probability density function. It's a function where you can think of the x values as the range of possible occurring values and y values as their probability of occurrence.

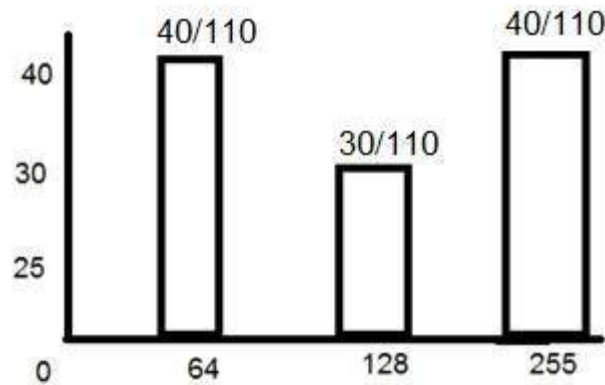
$$p(r_k) = \frac{n_k}{MN} \quad k = 0, 1, 2, \dots, L - 1$$

n_k : the number of pixels in the image of size $M \times N$ with intensity r_k

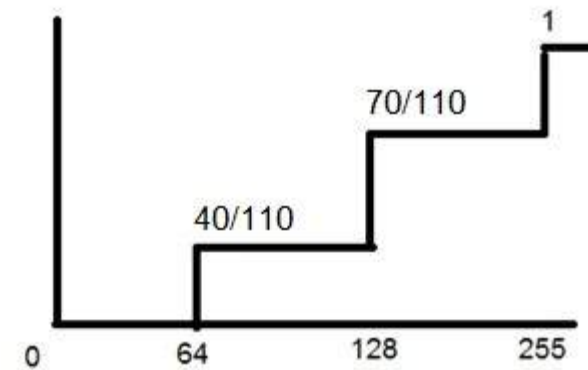


CDF

- **CDF** holds the probability of a probability distribution less than or equal to a particular value.
- It is a function that calculates the cumulative sum of all the values that are calculated by PDF.



PDF



CDF

Histogram Equalization

1. First calculate the PDF of all the pixels in this image.

$$p(r_k) = \frac{n_k}{MN} \quad k = 0, 1, 2, \dots, L-1$$

2. Calculate CDF (cumulative distributive function) of the pixels.
3. Multiply CDF of each pixel value with the highest intensity to satisfy the transformation function.

$$s_k = T(r_k) = (L-1) \sum_{j=0}^k p_r(r_j)$$

$$s_k = \frac{L-1}{MN} \sum_{j=0}^k n_j$$

4. The output image is obtained by mapping each pixel in the input image with intensity r_k into the corresponding pixel with level s_k in the output image.

Map s_k to its r_k

r_k	n_k	$p_r(r_k) = n_k/MN$	s_k
$r_0 = 0$	790	0.19	$S_0 = 1.33 = 1$
$r_1 = 1$	1023	0.25	$S_1 = 3.08 = 3$
$r_2 = 2$	850	0.21	$S_2 = 4.55 = 5$
$r_3 = 3$	656	0.16	$S_3 = 5.67 = 6$
$r_4 = 4$	329	0.08	$S_4 = 6.23 = 6$
$r_5 = 5$	245	0.06	$S_5 = 6.65 = 7$
$r_6 = 6$	122	0.03	$S_6 = 6.86 = 7$
$r_7 = 7$	81	0.02	$S_7 = 7.00 = 7$

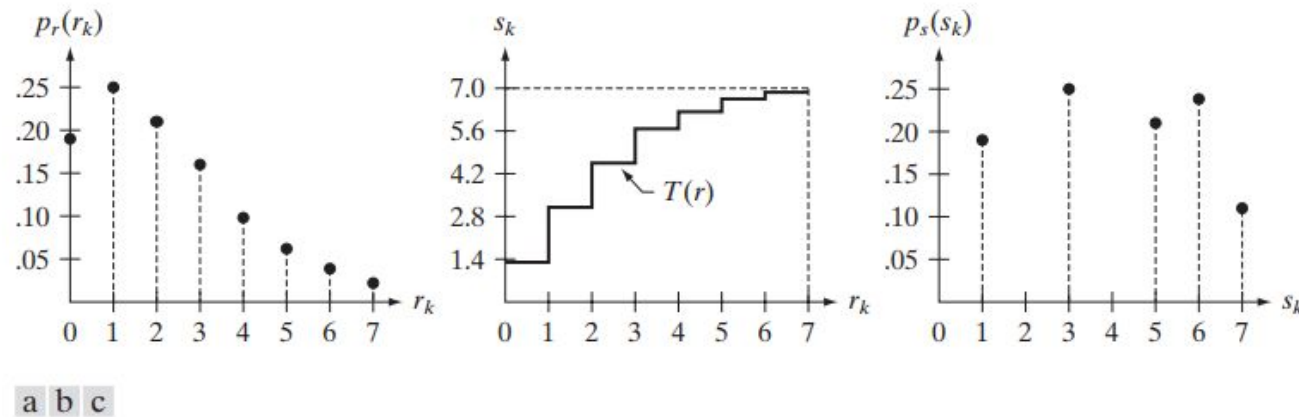


FIGURE 3.19 Illustration of histogram equalization of a 3-bit (8 intensity levels) image. (a) Original histogram. (b) Transformation function. (c) Equalized histogram.

Classwork

Take an input gray scale image and apply histogram equalization to enhance contrast:

1. Apply Histogram Equalization

a) `HE1 = Manual_equalizeHist(input_image)`

b) `HE2 = cv2.equalizeHist(input_img)`

2. Show the following outputs:

Input Image	Output Images (HE1 and HE2)
Input Histogram	Histogram of HE1
Input CDF	CDF of HE1

Python Functions

Using openCV:

```
cv2.calcHist(images, channels, mask, histSize,  
             ranges[, hist[, accumulate]])
```

```
hist_HE1 = cv2.calcHist([HE1],[0],None,[256],[0,256])
```

```
output = cv2.LUT(input_image, array)
```

- Show output Using Matplotlib:

```
plt.figure(figsize=(12,16))  
plt.subplot(3,2,1)  
plt.imshow(img, cmap='gray')  
plt.title("Original Image")  
plt.axis("off")
```

Show Outputs

```
plt.figure(figsize=(12,16))
```

```
# Row 1: Images
```

```
plt.subplot(3,2,1)
```

```
plt.imshow(img, cmap='gray')
```

```
plt.title("Original Image")
```

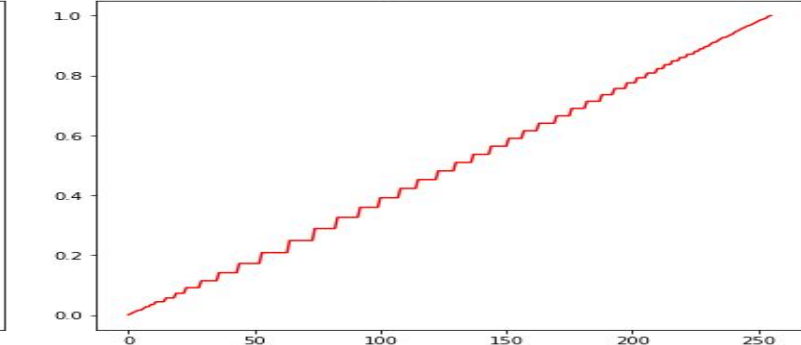
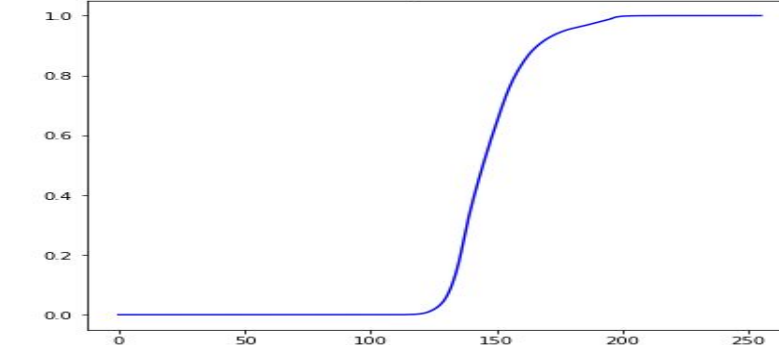
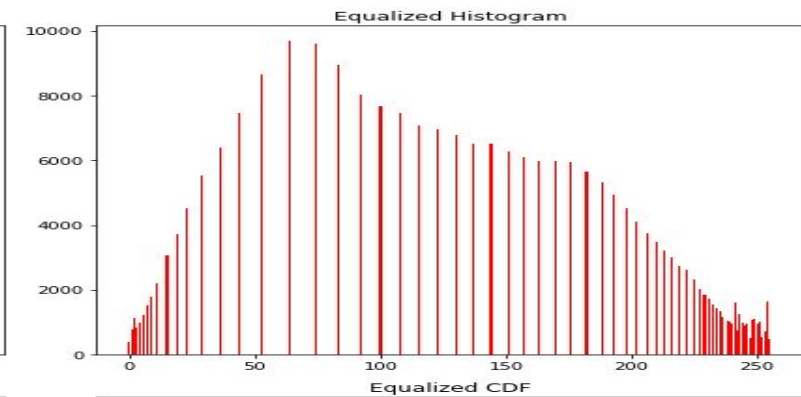
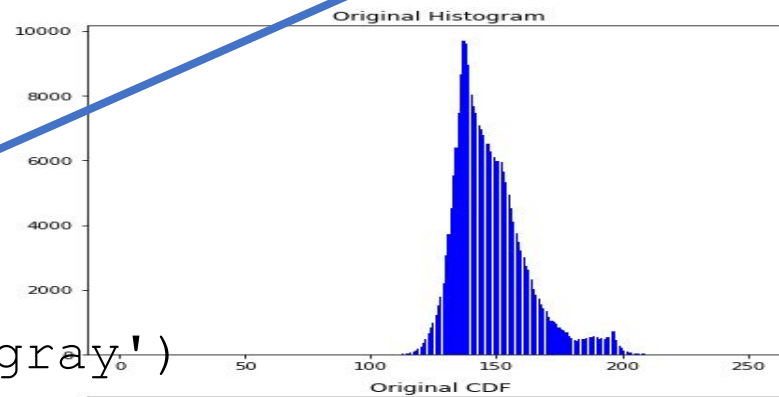
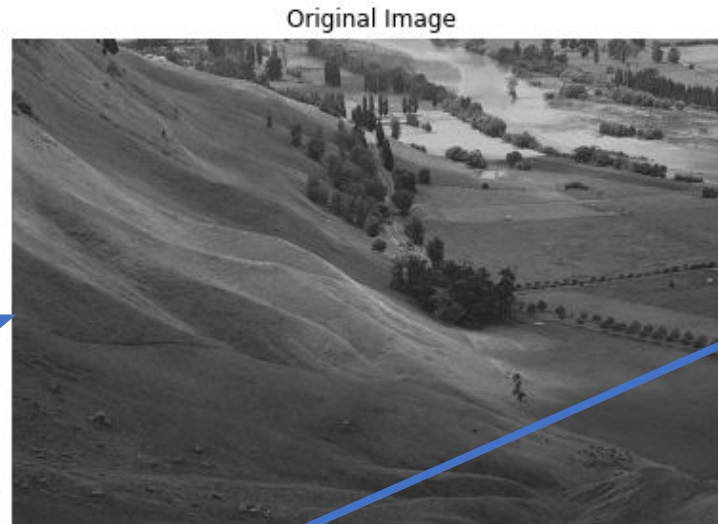
```
plt.axis("off")
```

```
plt.subplot(3,2,2)
```

```
plt.imshow(equalized, cmap='gray')
```

```
plt.title("Equalized Image  
(Manual)")
```

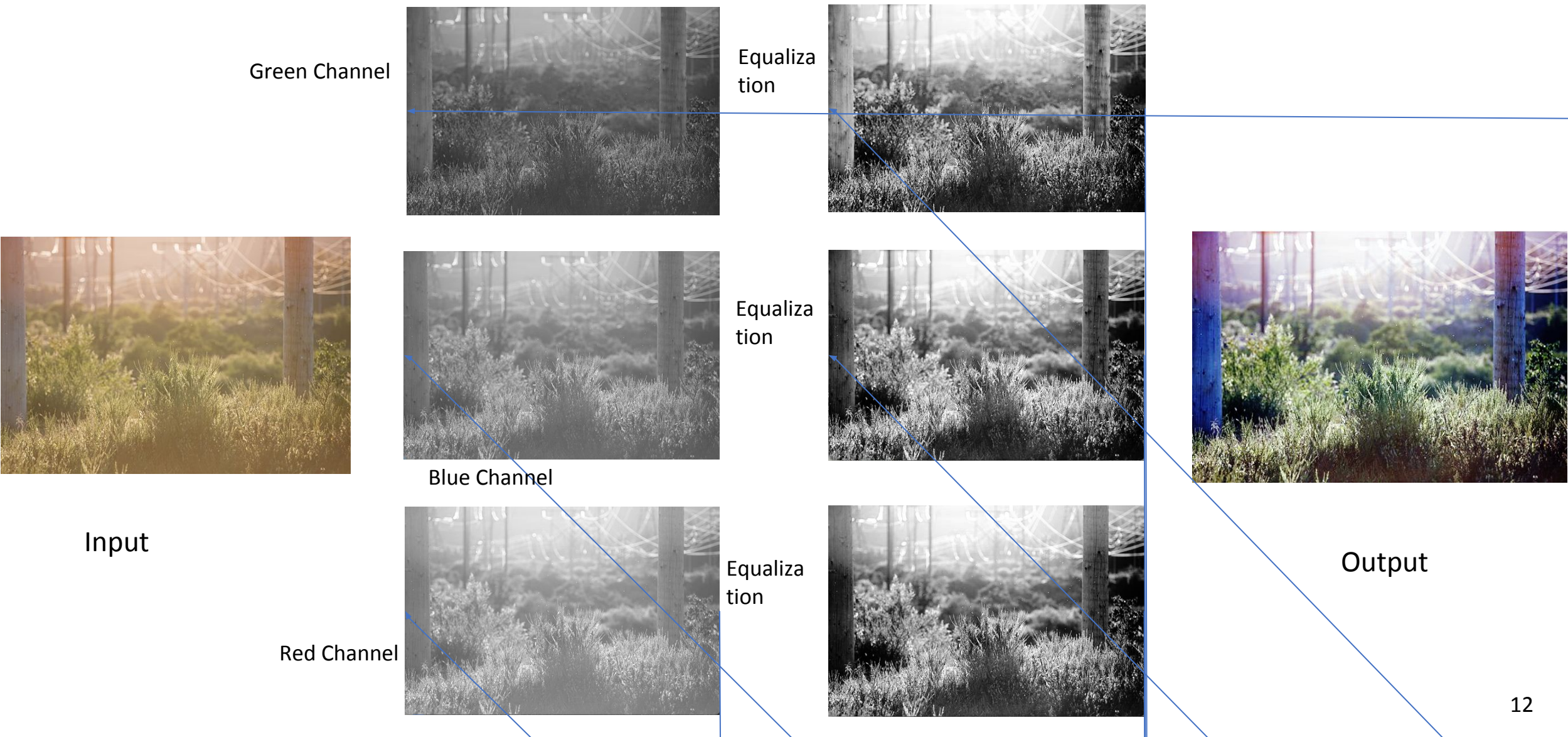
```
plt.axis("off")
```



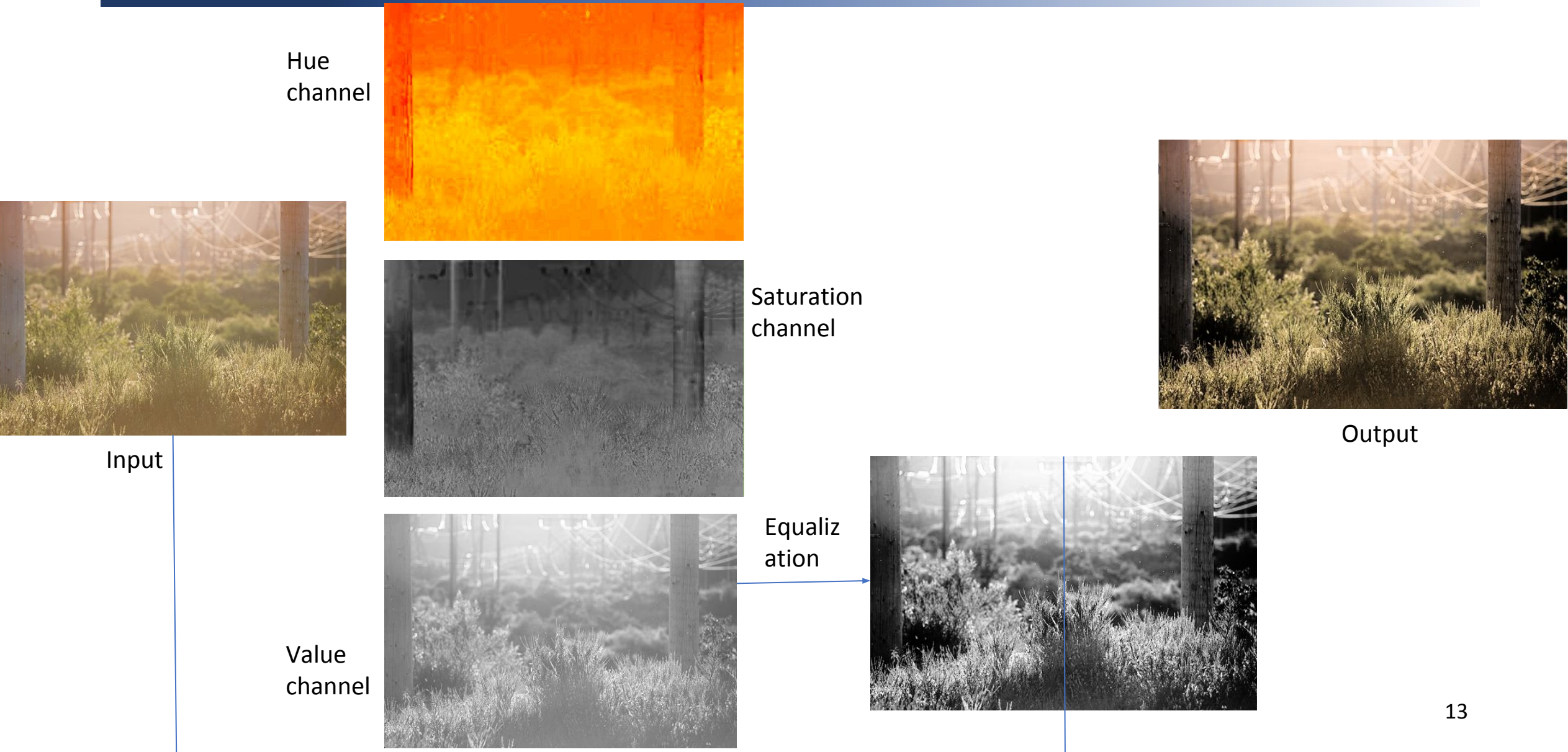
Assignment 1

1. Take an input color image and apply histogram equalization to enhance contrast.
 1. For the RGB image, apply histogram equalization to each of the 3 channels separately. Merge the 3 channels to view the enhanced image.
 2. Convert the RGB image to the HSV image and apply histogram equalization only in the Value channel. Merge the new equalized value channel with the other two to generate the final output equalized image.
2. Show the histogram of input and the equalized image
3. Show the histogram of CDF and PDF calculated from the input and output images.

Assignment 1



Assignment 1



Assignment 2

Histogram Matching (Specification)

-- where the target histogram follows an **Erlang distribution**.

Instructions:

1. Generate the target histogram using Erlang distribution and plot it for visualization. Take shape parameter (k) and scale parameter (μ) as user input.
2. Apply histogram matching to update the histogram of an input grayscale image to the target one.
3. Show the input and output image.
4. Plot the Histogram, PDF, and CDF of both input and output images.

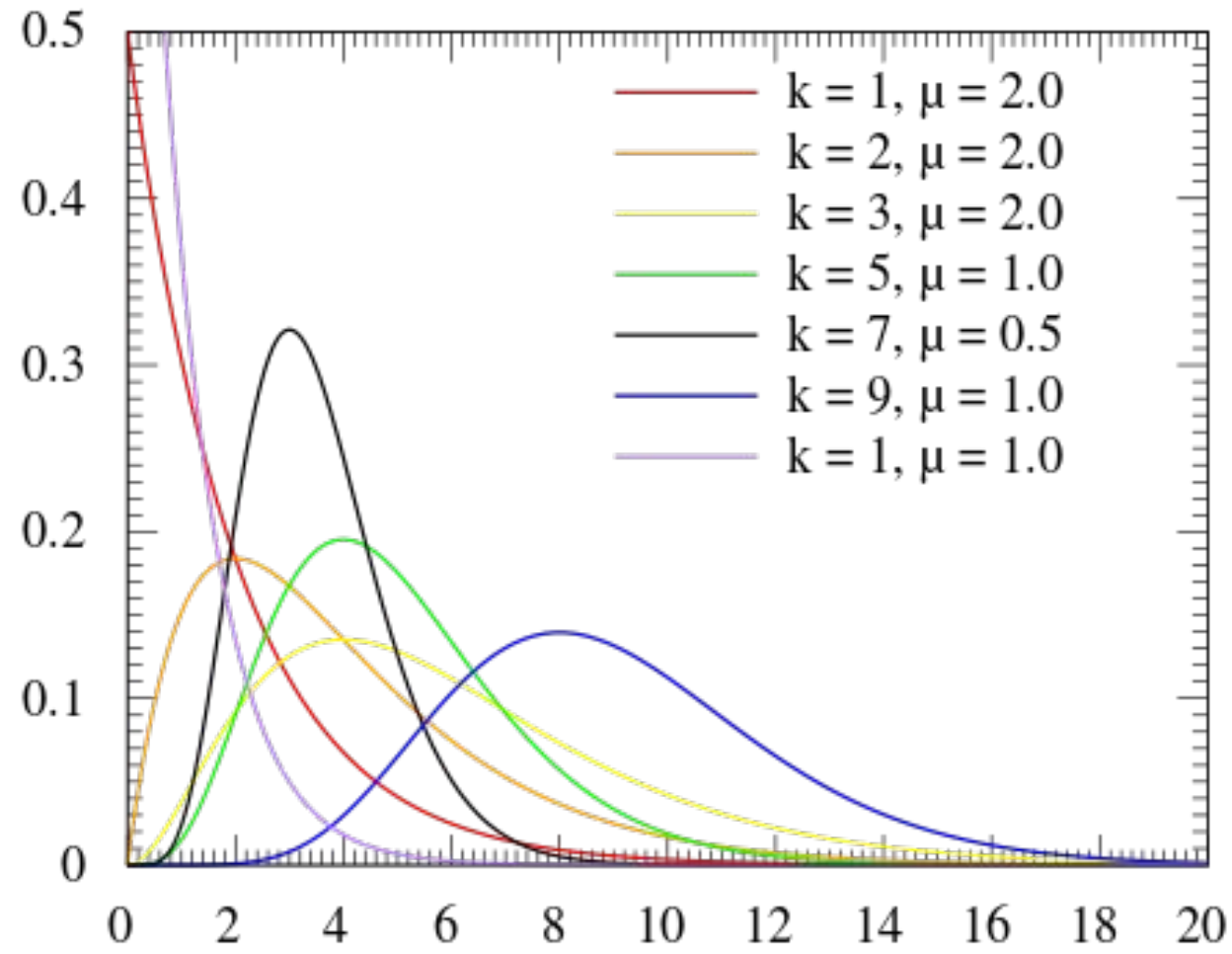
Erlang distribution

The Erlang distribution is a two-parameter family of continuous probability distributions

- Shape parameter (k) (positive integer)
- Rate parameter (λ) (positive real number)
 - The 'scale parameter', μ the reciprocal of the rate parameter, $\mu = 1/\lambda$.

$$f(x; k, \mu) = \frac{x^{k-1} e^{-\frac{x}{\mu}}}{\mu^k (k-1)!} \quad \text{for } x, \mu \geq 0.$$

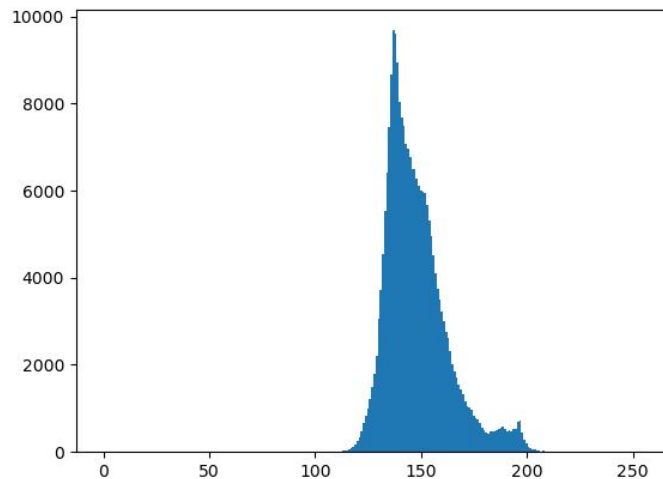
Erlang distribution



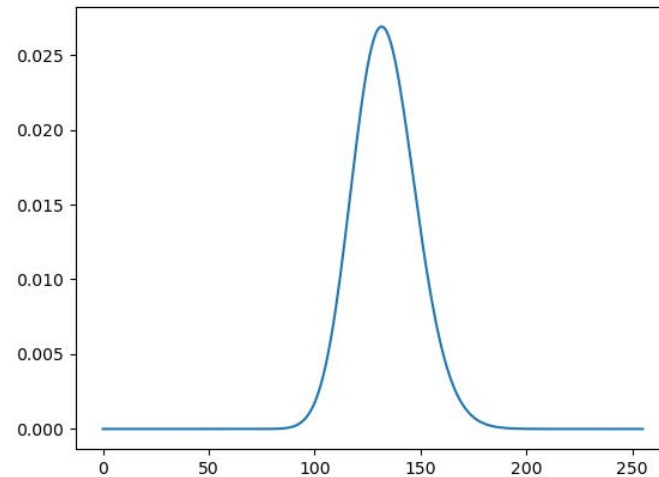
Histogram Matching with Erlang distribution



Input Image



Histogram of the Input Image



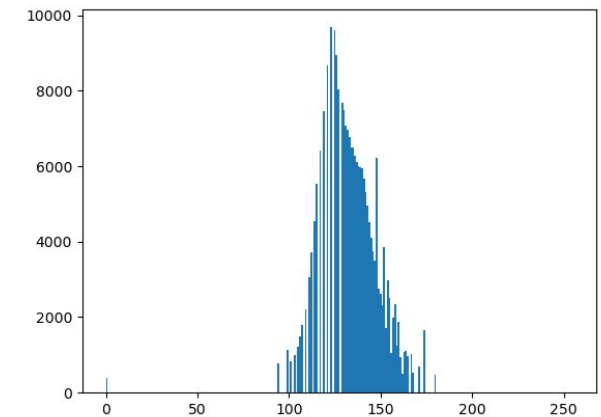
Target histogram

$$k = 80$$

$$\lambda = 0.6$$



Output Image



Histogram of the Output Image